



NEXT-GEN RURAL AI INFRASTRUCTURE

GramSetu AI — Zero to Master Guide

Complete technical, architectural, and presentation breakdown of an edge-computing, 100% offline digital kiosk for rural Bharat.

What is GramSetu AI & Why Was It Built?



The Core Real-World Crisis

Over 60% of rural India experiences poor, unstable 2G/3G connectivity or zero internet access. Farmers, rural women, and students miss out on critical government benefits (PM-Kisan, Ayushman Bharat, Crop Insurance) because current AI assistants (ChatGPT, Gemini) require high-speed cloud connectivity.

GramSetu AI is an autonomous, on-device AI Kiosk designed to run locally on low-power hardware (laptops, mini-PCs, or Raspberry Pi) placed in Village Panchayats, CSCs (Common Service Centers), and rural health centers.



100% Offline Edge Computing

Zero internet required. Operates completely offline with 0 cloud API fees and 0 network latency.



Zero Hallucination Guarantee

Uses a Deterministic SQLite Memory engine so critical rules (₹6,000/yr for PM-Kisan) are never hallucinated.



Local Language & Voice Native

Native voice interaction supporting Hindi, English, and Hinglish for non-literate villagers.



High Energy Efficiency

Consumes under 15W of power, allowing it to run for hours on solar panels or basic backup batteries.



What is AI, LLM & What Does "3B" Mean?



AI for Absolute Beginners

Artificial Intelligence (AI) is simply software trained to simulate human cognition — understanding natural language, pattern recognition, and decision making.

1. What is an LLM (Large Language Model)?

An LLM is a deep neural network trained on vast amounts of multilingual text. It predicts the most mathematically accurate and contextual next words to answer complex questions.

2. What Does "3B" Mean in Qwen 2.5 3B?

The "3B" stands for 3 Billion Parameters (~300 Crore digital synapses/weights).

Why 3B is the Perfect Sweet Spot:

- **0.5B / 1.5B Models:** Very fast, but weak logic. Prone to hallucinating government policy rules.
- **70B / 405B Models:** Highly intelligent, but require enterprise cloud GPUs costing ₹5,00,000+ and 40GB+ VRAM.
- **3B Model (Our Choice):** Requires only ~1.9 GB of RAM. Highly coherent in Hindi/Hinglish reasoning, runs at 25-40 tokens/sec on standard consumer CPU/RAM.

What is a Server & API? (The Restaurant Analogy)



1. Client (The Diner / Customer)

2. API (The Waiter)

3. Server (The Kitchen & Chefs)

4. Database (The Storage Pantry)

[illegible]



Programming Languages & Technologies Used

Technology	Role in GramSetu AI	Why It Was Chosen
Python 3.11	Backend logic & AI agent orchestration (<code>server.py</code> , <code>agent_decision.py</code>)	Rich ecosystem for AI, SQLite integration, and rapid edge deployment.
FastAPI	Asynchronous REST API framework	Extremely high performance, auto validation, and sub-millisecond routing.
Flutter & Dart	Cross-platform mobile application (<code>mobile_app/</code>)	Single codebase compiles to high-performance native Android APK and iOS.
C++ (llama.cpp)	Underlying Ollama AI execution engine	Ultra-fast matrix math computation directly on CPU registers without external GPU.
HTML5 / CSS3 / JS	Interactive Kiosk Web PWA (<code>index.html</code> , <code>styles.css</code>)	Lightweight, responsive Glassmorphism UI running smoothly on any browser.
SQLite3	Embedded offline relational database (<code>gramsetu.db</code>)	Zero-configuration, lightweight, ACID-compliant local database.

Prompting & Hyperparameter Tuning

Prompting is the science of designing precise instructions so the AI responds strictly within desired bounds without wandering off topic.

System Prompt (The Constitution)

Instructs the model: *"You are GramSetu AI, an offline digital assistant for rural citizens. Answer strictly in 2-3 bullet points using only verified local data."*

Temperature (Set to 0.15)

Controls creativity vs factuality. **0.0-0.2** forces the model to be strict, consistent, and factual. High values (0.8+) cause hallucinations.

Repeat Penalty (Set to 1.45)

Penalizes the model for repeating identical sentences, ensuring clean, concise answers on edge hardware.

Curated RAG (Open-Book Exam)

Injects verified database records directly into context so the AI reads from verified sources rather than guessing.



What is Google Antigravity & How Was It Used?

Google Antigravity Overview:

Google Antigravity is Google DeepMind's advanced autonomous agentic coding environment and pair programmer. It operates with deep repo-level reasoning, automated terminal tool execution, and self-testing capabilities.

How Antigravity Built GramSetu AI:

- **Architecture Design:** Architected the dual-tier system (Deterministic SQLite + Local LLM Reasoning).
- **Ollama & Qwen 2.5 3B Integration:** Automated the configuration, parameter tuning (Temperature 0.15), and prompt sanitization.
- **Flutter & Mobile App Scaffolding:** Structured the Flutter screens for Kisan, Swasthya, and Sarkari modules.
- **Automated Test Suite:** Developed and verified 10/10 test cases in `test_suite.py` ensuring zero hallucination on critical schemes.

Top Viva & Jury Questions & Answers

Q1: What happens if the internet goes down completely?

Answer: GramSetu AI is 100% offline. Both the Qwen 2.5 3B AI model and the SQLite database reside locally on the kiosk hardware. It operates continuously with zero connectivity.

Q2: How do you prevent the AI from giving wrong (hallucinated) medical or scheme advice?

Answer: We employ a hybrid architecture. Critical data (PM-Kisan installment amounts, emergency helpline numbers) is fetched deterministically from SQLite and passed as RAG ground-truth context to the LLM with Temperature 0.15.

Q3: Why not use cloud-based APIs like OpenAI GPT-4 or Google Gemini?

Answer: Cloud APIs require constant internet, incur recurring per-token subscription costs, and have high latency. GramSetu AI has 0 operational cost after deployment and protects citizen data locally.

Q4: How does a non-literate villager interact with the system?

Answer: The interface provides single-tap voice recording with multilingual Speech-to-Text and Text-to-Speech (Hindi/English), speaking answers aloud clearly.